Example 9.20

The following data gives the readings for 8 samples of size 6 each in the production of a certain product. Find the control limits using mean chart.

Sample	1	2	3	4	5	6
Mean	300	342	351	319	326	333
Range	25	37	20	28	30	22

Given for $n = 6$, $A_2 = 0.483$,

Solution:

Sample	1	2	3	4	5	6	**Total**
Mean	300	342	351	319	326	333	**1971**
Range	25	37	20	28	30	22	**162**

Table 9.20

$$\overline{\overline{X}} = \frac{\sum \overline{X}}{number\ of\ samples} = \frac{1971}{6} = 328.5 \qquad \overline{R} = \frac{\sum R}{n} = \frac{162}{6} = 27$$

The control limits for $\overline{X}$ chart is

$$UCL = \overline{\overline{X}} + A_2\overline{R} = 328.5 + 0.483(27) = 341.54$$

$$CL = \overline{\overline{X}} = 328.5$$

$$LCL = \overline{\overline{X}} - A_2\overline{R} = 328.5 - 0.483(27) = 315.45$$

Example 9.21

The data shows the sample mean and range for 10 samples for size 5 each. Find the control limits for mean chart and range chart.

Sample	1	2	3	4	5	6	7	8	9	10
Mean	21	26	23	18	19	15	14	20	16	10
Range	5	6	9	7	4	6	8	9	4	7

Solution:

Sample	1	2	3	4	5	6	7	8	9	10	**Total**
Mean	21	26	23	18	19	15	14	20	16	10	**182**
Range	5	6	9	7	4	6	8	9	4	7	**65**

Table 9.21

$$\overline{\overline{X}} = \frac{\sum \overline{X}}{number\ of\ samples} = \frac{182}{10} = 18.2 \qquad \overline{R} = \frac{\sum R}{n} = \frac{65}{10} = 6.5$$

XII Std - Business Maths & Stat EM Chapter 9.indd 224 2/28/2020 5:11:30 PM